Annexes

Annexe 1

	brand	id	mois	prot	fat	ash	sodium	carb	cal
26	A	34006	28.93	19.99	45.20	4.78	1.62	1.10	4.91
27	A	24146	30.41	18.71	43.99	4.86	1.62	2.03	4.79
28	A	24138	29.62	21.10	43.37	5.05	1.69	0.86	4.78
29	B	14015	49.57	13.70	29.07	3.62	1.10	4.04	3.33
30	B	14006	52.68	14.38	25.72	3.26	0.93	3.96	3.05
31	B	14024	48.53	13.14	30.38	3.55	0.99	4.40	3.44

Figure 1: extrait des données

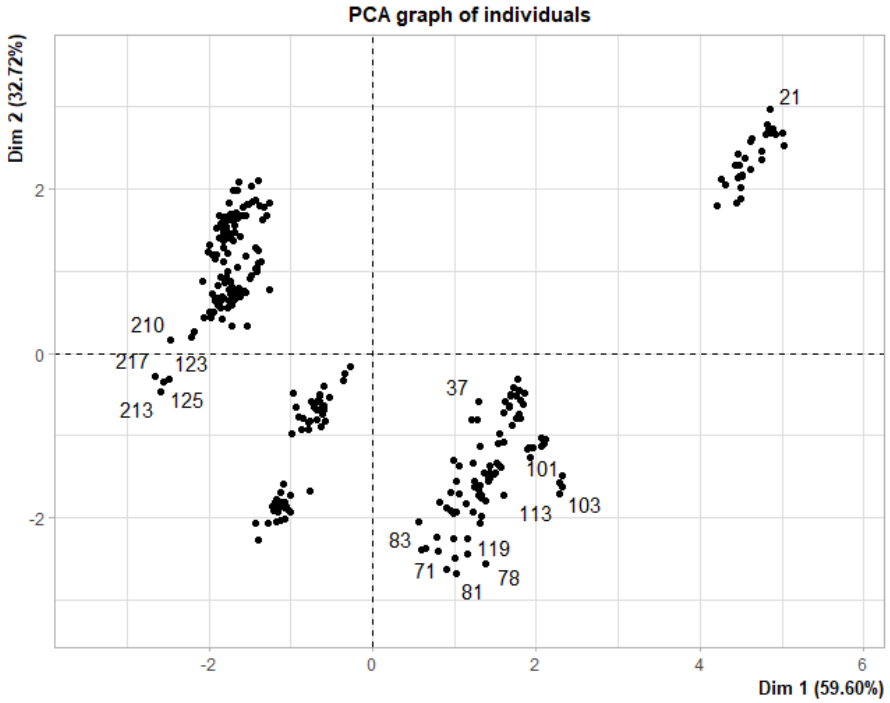


Figure 2: carte des individus

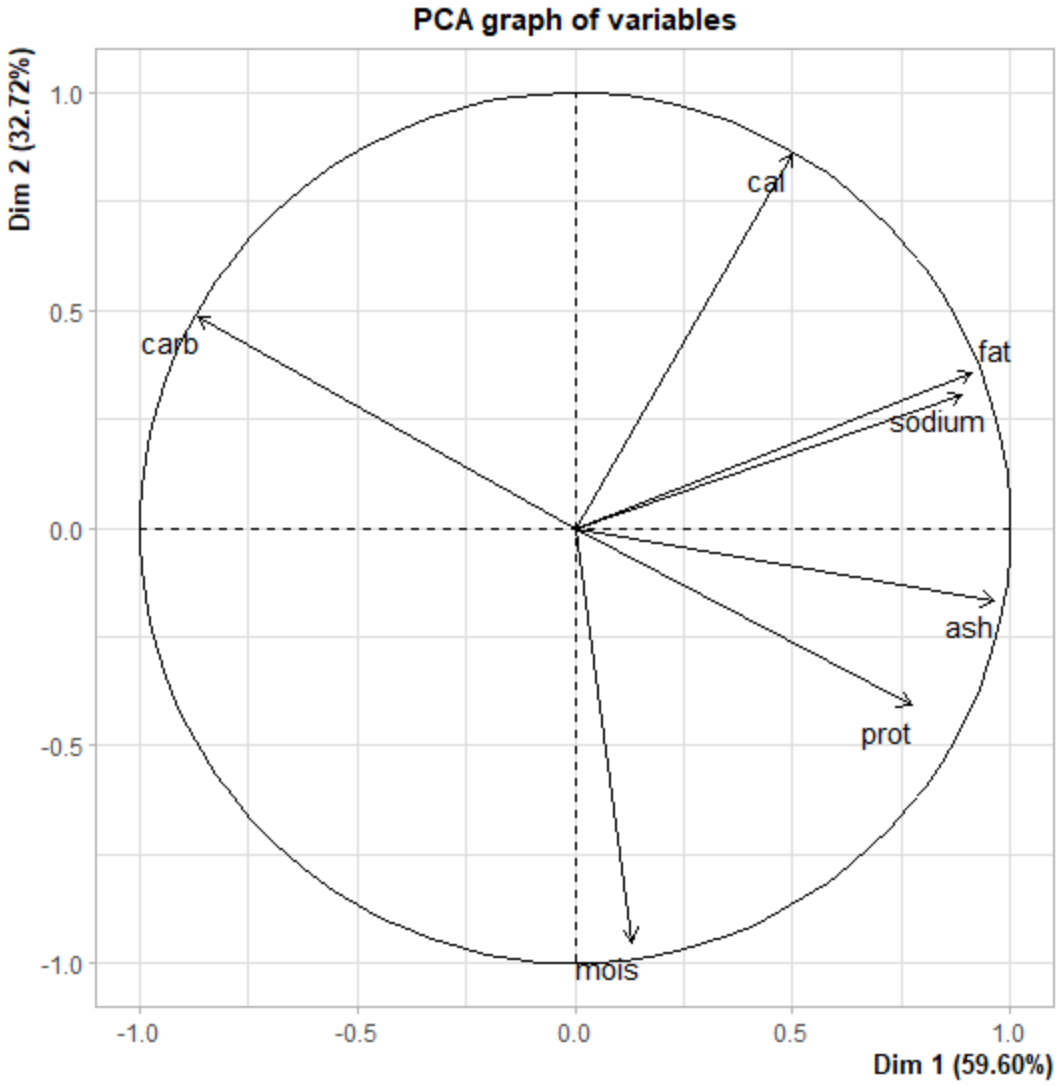


Figure 3: Cercle de corrélation

Annexe 2

Mois	Dépenses TV	Dépenses Radio	Dépenses Journaux	Ventes
1	5000	1000	500	55000
2	7000	2000	700	67000
3	5500	1500	600	62000
4	4000	1000	400	55000
5	6500	1800	900	70000
6	8000	2300	1000	82000
7	3500	800	300	42000
8	5000	1200	500	55000
9	7000	2100	800	71000
10	6000	1900	700	68000
11	4500	1300	500	59000
12	5500	1500	600	61000

Annexe 3

	Serial.No.	GRE.Score	TOEFL.Score	University.Rating	SOP	LOR	CGPA	Research	Chance.of.Admit
1	1	337	118	4	4.5	4.5	9.65	1	0.92
2	2	324	107	4	4	4.5	8.87	1	0.76
3	3	316	104	3	3	3.5	8	1	0.72
4	4	322	110	3	3.5	2.5	8.67	1	0.8
5	5	314	103	2	2	3	8.21	0	0.65
6	6	330	115	5	4.5	3	9.34	1	0.9
7	7	321	109	3	3	4	8.2	1	0.75
8	8	308	101	2	3	4	7.9	0	0.68
9	9	302	102	1	2	1.5	8	0	0.5
10	10	323	108	3	3.5	3	8.6	0	0.45

Figure 1: Extrait des données

```
> summary(x)
   Serial.No.      GRE.Score      TOEFL.Score    University.Rating      SOP
 Min.   :  1.0   Min.   :290.0   Min.   : 92.0   Min.   :1.000     Min.   :1.0
 1st Qu.:100.8   1st Qu.:308.0   1st Qu.:103.0   1st Qu.:2.000     1st Qu.:2.5
 Median :200.5   Median :317.0   Median :107.0   Median :3.000     Median :3.5
 Mean   :200.5   Mean   :316.8   Mean   :107.4   Mean   :3.087     Mean   :3.4
 3rd Qu.:300.2   3rd Qu.:325.0   3rd Qu.:112.0   3rd Qu.:4.000     3rd Qu.:4.0
 Max.   :400.0   Max.   :340.0   Max.   :120.0   Max.   :5.000     Max.   :5.0
      LOR             CGPA          Research      Chance.of.Admit
 Min.   :1.000   Min.   :6.800   Min.   :0.0000   Min.   :0.3400
 1st Qu.:3.000   1st Qu.:8.170   1st Qu.:0.0000   1st Qu.:0.6400
 Median :3.500   Median :8.610   Median :1.0000   Median :0.7300
 Mean   :3.453   Mean   :8.599   Mean   :0.5475   Mean   :0.7244
 3rd Qu.:4.000   3rd Qu.:9.062   3rd Qu.:1.0000   3rd Qu.:0.8300
 Max.   :5.000   Max.   :9.920   Max.   :1.0000   Max.   :0.9700
```

Figure 2: Résultat de la fonction Summary

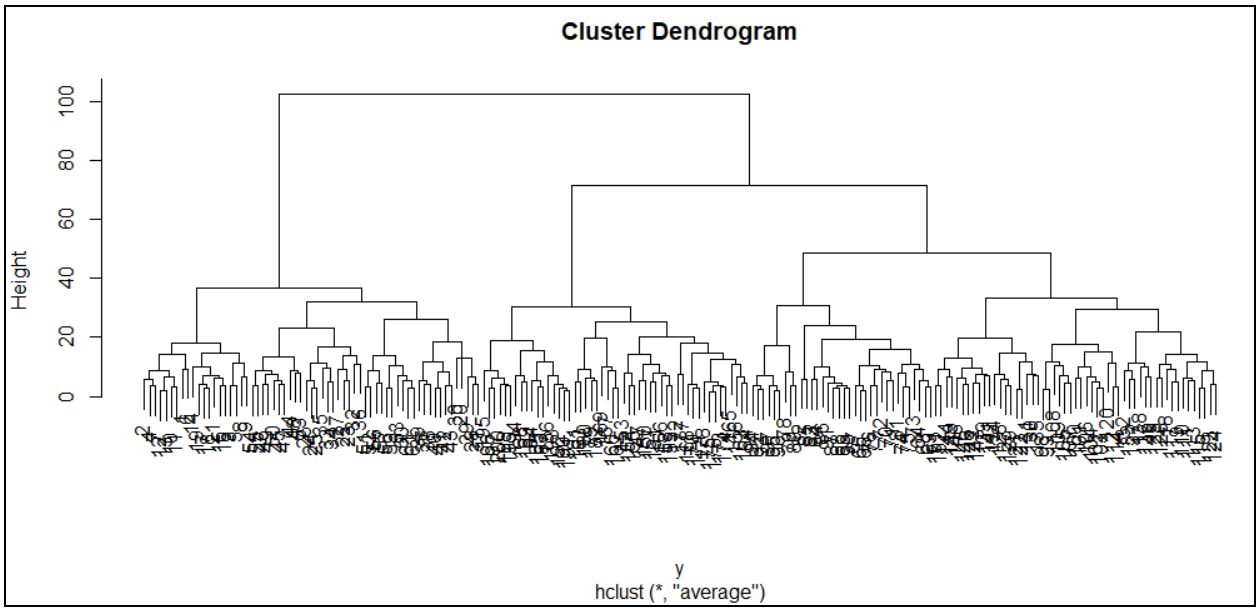


Figure 3: Dendrogramme

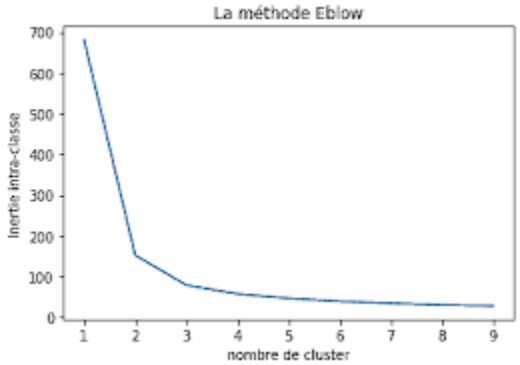


Figure4: Elbow visualization

```
1) root 400 191 non (0.52250000 0.47750000)
  2) CGPA< 8.735 234  38 non (0.83760684 0.16239316)
    4) GRE.Score< 318.5 195  18 non (0.90769231 0.09230769) *
    5) GRE.Score>=318.5 39  19 oui (0.48717949 0.51282051)
     10) GRE.Score< 326 32  14 non (0.56250000 0.43750000)
       20) LOR< 4.25 29  11 non (0.62068966 0.37931034)
         40) LOR>=2.75 21   5 non (0.76190476 0.23809524) *
         41) LOR< 2.75 8   2 oui (0.25000000 0.75000000)
           82) GRE.Score>=323 2   0 non (1.00000000 0.00000000) *
           83) GRE.Score< 323 6   0 oui (0.00000000 1.00000000) *
       21) LOR>=4.25 3   0 oui (0.00000000 1.00000000) *
     11) GRE.Score>=326 7   1 oui (0.14285714 0.85714286) *
  3) CGPA>=8.735 166  13 oui (0.07831325 0.92168675) *
```

Figure 5: Résultat textuel de l'arbre de décisions